Dealing with Missing Data

Machine Learning DSA-447/DGE-301

What is Missing Data

Often when data is collected, there are some missing values apparent in the dataset. This leads to a few questions to consider:

- How does this show up in pandas?
- How does sklearn handles these NaNs?
- How does this effect our modeling?
- What are the simplest ways to handle missing data?

  - A general pattern describes missing values that are dispersed throughout the data matrix.
  - Missingness may or may not be systematic.

X1	X2	Y2	Y2
?		?	
?	?		
?			?
		?	?
		?	
?			?
?			
	?		
		?	

Missing Data

Missing Data Mechanisms

- Mechanisms describe how the probability of a missing value Y relates to other variables or to the would-be values of Y itself.

  - MCAR: Missing Completely at Random.
    - Missingness is not related to any other variable (e.g., Random system glitch)
  - MAR: Missing at Random.
    - Missingness depends on other observed data (e.g., small companies do not report salaries)
  - MNAR: Missing No at Random.
    - Missingness depends on the missing values itself (e.g., high-income people refuse to disclose salary.

How Missing Data Affects Models

- Training stops if dataset contains NANs (this does not fit for all ML algorithms)
- Causes biased estimates
- Reduces model accuracy.
- Affect distributions (mean, variance)
- Impact visualizations.
- Can produce wrong prediction.

X	Y
1	2
2	NaN
3	6

model.fit(X, Y)

```
ValueError: Input contains NaN
```

Person	Income
A	2000
B	NaN
C	NaN
D	5000

Average income **without missing handling**:

$$\frac{2000 + 5000}{4} = 1750$$

But this is wrong because missing ≠ zero.

Correct average (excluding NaN):

$$\frac{2000 + 5000}{2} = 3500$$

The result **looks poorer** than the population actually is → **biased**.

Age	Salary
25	40k
NaN	50k
35	60k

NaN imputed as mean = 30

But real missing age might be **50** → salary relation is wrong.

Impact:

- Model learns an incorrect pattern
- Predictions become less accurate

When Should we Drop Missing Data?

- Drop Rows When:
  - Only few rows missing
  - Missingness is random
  - Dataset is large and loss is acceptable.
- Avoid Dropping When:
  - Dataset is small.
  - Missingness is informative
  - Column is important for prediction.

- Drop Columns When:
  - 50-70% values missing
  - Column is not important
  - No reliable imputation exists

Missingness is **informative** when:

Medical Test Results

Patient	Age	Glucose Level
A	25	90
B	55	110
C	72	NaN
D	65	95

$P(\text{Missing})$ depends on X

Missing = possible high glucose
This means the missingness itself contains information.

Why is Patient C's value missing?

- Some patients with very high blood sugar skip tests.
- Fear of the result
- Already diagnosed; test not taken
- Hospitalized before testing

Meaning of Missingness

- The missing value may indicate high risk
- Missingness is not random
- Missing = valuable signal, not noise

Imputation Overview

- **What is Imputation?**
  - Filling missing values with estimated or replacement values.
  - Imputation methods:
    1. Mean imputation (numeric)
    2. Median imputation (numeric, skewed data)
    3. Mode imputation (categorical)
    4. Constant value imputation
    5. Group-wise imputation (industry-wise, location-wise)
    6. Advanced methods (KNN, regression)

Mean Imputation:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Replace each missing value with the computed mean.

When to use:

- Numeric data
- Data roughly symmetric (not skewed)
- No extreme outliers

$$[50, 55, \text{NaN}, 60]$$

$$\downarrow$$

$$\frac{50+55+60}{3} = 55 \longrightarrow [50, 55, 55, 60]$$

Imputation Overview

- **Median Imputation**

Median = middle value in sorted data

When to use:

- Skewed numeric data
- Outlier presence (salary, revenue)

Example:

$$[10, 12, 11, 100, \text{NaN}]$$

Median = 12

Replace NaN → 12

- **Mode Imputation (Categorical)**

Example:

`['Tech', 'Tech', 'Finance', NaN]`

Mode = `'Tech'`

When to use:

- Categorical data
- Discrete labels

- **Constant Value Imputation**

Examples:

- Fill missing Industry → `"Unknown"`
- Fill missing Rating → `0`
- Fill missing Salary → `"Not Provided"`

Good for:

- Features where "missing" has meaning
- Textual fields
- Survey data

- **Group-Wise Imputation**

Impute missing value based on similar groups.

Example:

Missing `"Rating"`

→ Replace with median rating **within the same Industry.**

Dealing with Outliers

Outlier

Outlier

A data point that is **significantly different** from the rest of the dataset.

Characteristics:

- Unusually high or low values
- Do not follow the general pattern
- May indicate errors or meaningful rare events

Bad Outliers

- Errors
- Typos
- Sensor malfunction
- Wrong scale (e.g., salary = 30000 vs 300000)

Useful Outliers

- Fraud detection
- Medical abnormalities
- Rare events
- High-value customers

Example 1:

Dataset:

`10, 12, 13, 11, 12, 95`

→ **95** is an outlier

Example 2:

Dataset:

`5, 6, 7, 8, 100`

→ **100** is an outlier

Example 3:

Dataset:

`60, 62, 63, 65, 67, -10`

→ **-10** is an impossible outlier (measurement error)

Outlier Detection Methods

- **Z-Score**
  - Measure how far a value is from the mean in terms of standard deviation.

$$Z = \frac{x - \mu}{\sigma}$$

Rule:

- |Z| > 3 ⇒ Outlier

Dataset:

`10, 12, 11, 50`

Mean = 20.75

Std Dev ≈ 18.2

Compute Z for 50:

$$Z = \frac{50 - 20.75}{18.2} \approx 1.6$$

50 is **NOT** an outlier under Z-score (dataset too small).

Use when:

- Data is **normally distributed**
- No extreme skew
- Large enough dataset (>30 points)

Avoid when:

- Skewed distributions
- Salaries, prices, revenues

Outlier Detection Methods

- **IQR**
  - Interquartile Range (IQR).

$$IQR = Q3 - Q1$$

  - Outlier boundaries:

$$\text{Lower} = Q1 - 1.5 \cdot IQR$$

$$\text{Upper} = Q3 + 1.5 \cdot IQR$$

Values outside → **outliers**

Example: $X = [11, 10, 12, 12, 13, 40]$ **(1)** Sort → $X = [10, 11, 12, 12, 13, 40]$

(2) Find the Median (Q2) $Q_2 = \frac{12+12}{2} = 12$

(3) Split the Data Into Two Halves: lower half (10,11,12), upper half (12,13,40)

(4) Compute Q1 (First Quartile):

Q1 = **median of the lower half**

Lower half = $[10,\ 11,\ 12]$

This has 3 values → median = middle value = **11**

$$Q1 = 11$$

(5) Compute Q3 (Third Quartile)

Q3 = **median of the upper half**

Upper half = $[12,\ 13,\ 40]$

This also has 3 values → median = middle value = **13**

$$Q3 = 13$$

(6) Compute IQR

$$IQR = Q3 - Q1$$

$$IQR = 13 - 11 = 2$$

(7) Compute Outlier Boundaries

Lower Bound:

$$\text{Lower Bound} = Q1 - 1.5 \cdot IQR = 11 - 1.5(2) = 11 - 3 = 8$$

Upper Bound:

$$\text{Upper Bound} = Q3 + 1.5 \cdot IQR = 13 + 1.5(2) = 13 + 3 = 16$$

(8) Values > 16:
→ **40** is an outlier.